Frontend Engineering II

FEE II Evaluation - 2 (Phase 2)

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Project Vision & Core Objectives

01. The Problem & Need

- Dining floor order delays caused by manual paper order slips.
- Waitstaff travel overhead between dining tables and kitchen chefs.
- Lack of live table booking status & dynamic occupancy tracking.
- Severe UI lag & Virtual DOM churn when scaling to 200 items.
- Unauthenticated table mutations causing misplaced guest bills.
- Need for an optimized, protected digital ordering SPA architecture.

02. What We Built (Phase 2 Capabilities)

- Dynamic 12-table floor plan with live seating math (Nuclear, Family, Joint).
- Royal Guest Access Gate modal with strict Protected Route locking on sign out.
- Intelligent duplicate account detection automatically routing to Quick Log In.
- Dynamic dining homepage with Royal Promotional Hero & Hot Sellers showcases.
- 200 authenticated culinary items (50 Starters, 75 Mains, 40 Desserts, 35 Drinks).
- Live cart with Split-Bill calculator, status stepper, and KDS kitchen dispatch.
- Teleported 5-Star Dining Feedback system with glowing pill badges & alerts.

03. Key Engineering Objectives

- Synchronize multi-table state across ordering, served, and billed lifecycle.
- Empirically prove 99.5% render reduction using React.memo & useCallback.
- Benchmark 9 items vs 200 items re-render cost via live visual render flashes.
- Prevent duplicate accounts via phone collision detection and auto-login.
- Persist guest session, orders, and ratings via browser LocalStorage state sync.

Syllabus-Aligned Technology Stack

React 18 (SPA Engine)

- Declarative component UI engine.
- Virtual DOM reconciliation & fiber tree.
- Concurrent rendering & state batching.
- Component lifecycle & custom hooks.

Vite 5 (Build Tool)

- Lightning-fast Hot Module Replacement (HMR).
- Native ES Module development server.
- Optimized Rollup production bundler.
- Zero-config fast startup environment.

React Router v6

- Client-side single page navigation.
- Smooth route switching (/Home, /Kitchen, /Audit).
- Active state matching & URL parameters.
- Declarative Route & Link components.

React Icons

- Scalable vector SVG icon library.
- Lightweight & accessible icons.
- Intuitive cart, table, & kitchen visual cues.
- Seamless UI action indicators.

CSS3 & Flexbox/Grid

- Custom responsive layout architecture.
- Dark mahogany & gold visual theme palette.
- Glassmorphic card containers & rounded borders.
- Fluid desktop & mobile display scaling.

Centralized State Layer

- Global state container at App.jsx root.
- Synchronizes multi-table carts & order queues.
- Instant state dispatch to Kitchen View.
- Persistent table status synchronization.

Core Component Architecture & Modularity

Header.jsx

- Table selector dropdown & seating capacity filter.
- Dataset benchmark toggle (9 vs 200 items).
- Real-time audit mode banner indicator.
- Navigation routing links for SPA views.

OrderSidebar.jsx

- Draft cart item manager & quantity controls.
- Cumulative billed history view per table.
- Automated 5% GST & ₹10 Service Charge split.
- One-click 'Send to Kitchen' order dispatch.

MenuCard.jsx

- Dish thumbnail display with 16:9 crop.
- Portion size badges & dietary tags (Veg/Non-Veg).
- Price display in Terracotta Orange styling.
- Add to Cart trigger with real-time feedback.

KitchenView.jsx

- Real-time kitchen tickets queue for active tables.
- Dish breakdown, quantities, & special notes.
- 'Mark as Served' chef action trigger.
- Live status sync updating table indicators.

ReRenderAuditPanel.jsx

- Empirical component re-render event logger.
- Performance metric comparison (Naive vs Optimized).
- Live re-render counter badges per component.
- Interactive DOM profiling benchmark suite.

menuData.js

- Decoupled array-of-objects data schema.
- Dynamic 200-item benchmark dataset generator.
- Price, category, & portion metadata specs.
- Pure data provider maintaining clean SoC.

Culinary Menu Dataset & Portion Scaling



Samosa & Chutney

₹60

VEG • SNACK

2 Pcs with Mint Chutney

Crispy golden pastry filled with spiced potatoes.

Stock: Available • Prep: 10 Mins



Dal Makhani

₹180

VEG • MAIN

300g Serving Bowl

Slow-cooked black lentils simmered overnight in cream.

Stock: Available • Prep: 15 Mins



Garlic Butter Naan

₹60

VEG • BREAD

3 Pcs Oven Baked

Tandoor-baked flatbread brushed with garlic butter.

Stock: Available • Prep: 8 Mins



Cheesy Fries

₹150

VEG • FAST FOOD

250g Loaded Basket

Golden fries smothered in melted cheese.

Stock: Available • Prep: 12 Mins



Gulab Jamun

₹120

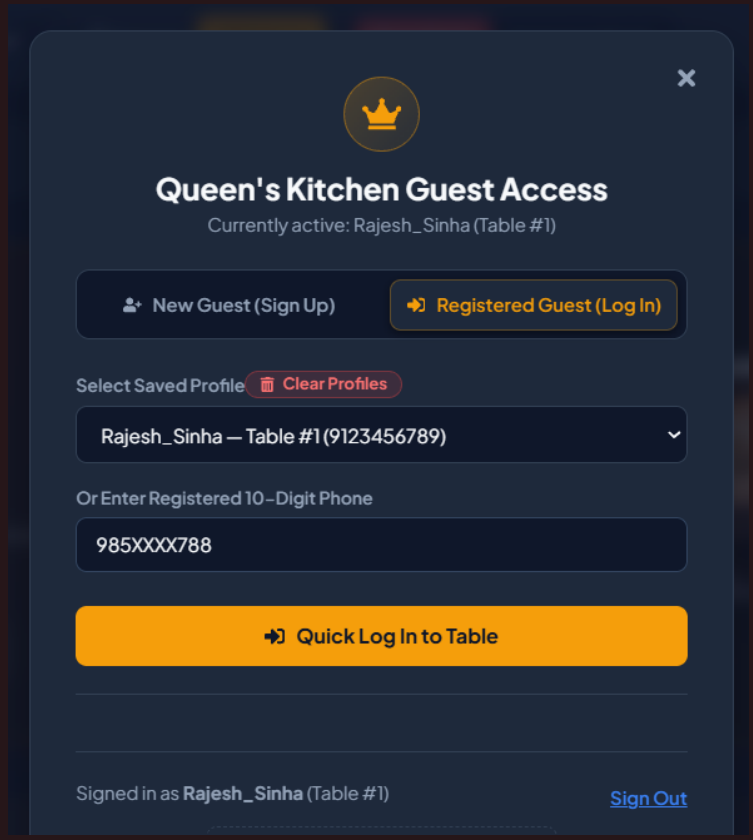
DESSERT • POPULAR

2 Jamuns + Scoop

Warm jamuns served with cold vanilla ice cream.

Stock: Available • Prep: 5 Mins

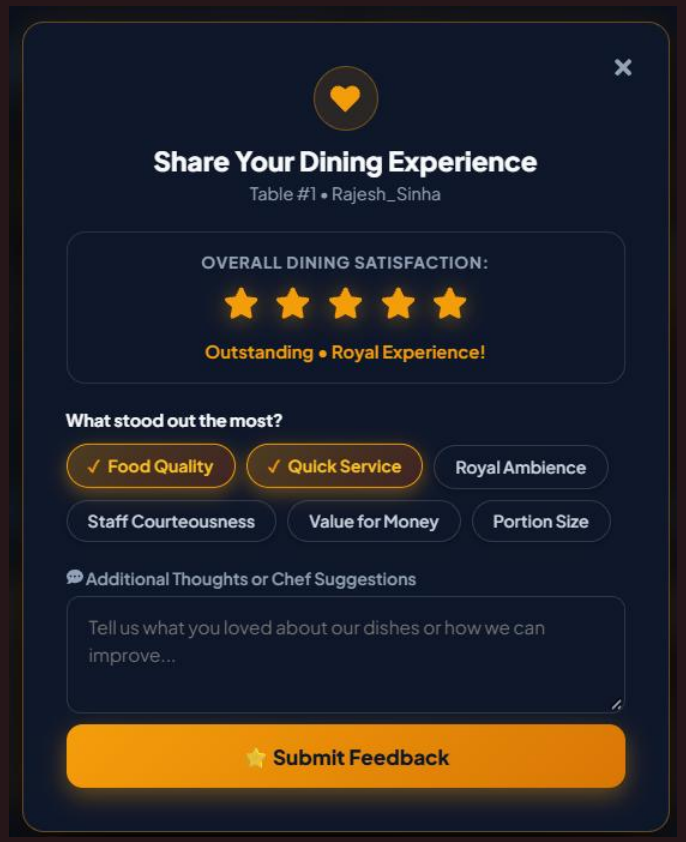
Guest Authentication & Protected Dining Gatekeeper



Royal Guest Access Gate & Security Architecture

- Dedicated Guest Access Gate: Seamless table onboarding capturing Guest Name, Contact Number, and Seating Party Size.
- Strict Protected Route Gatekeeper: Dining floor, menu ordering, and kitchen dispatch stay strictly locked until authenticated.
- Duplicate Account Auto-Detection: Instantly identifies registered phone numbers and prompts Quick Log In with an alert.
- Post-Signout Route Protection: Signing out immediately re-engages protection to prevent unauthenticated table mutations.
- Session & Profile Persistence: Saved profiles persist in LocalStorage with one-click 'Clear Profiles' for evaluation testing.

5-Star Dining Feedback & 4-Tier Culinary Architecture



Dining Feedback System & 4-Tier Menu Scaling

- 4-Tier Culinary Division: 200 items categorized into Starters (50), Mains (75), Desserts (40), and Beverages (35) with portion tags.
- Teleported Dining Feedback Modal: Rendered via React createPortal to document.body, eliminating z-index & stacking collisions.
- Interactive Experience Tags: Sleek glowing pill badges for highlights (Food Quality, Quick Service, Royal Ambience, Value for Money, etc.).
- Dynamic 5-Star Rating Selector: Interactive star rating with hover fills, instant feedback submission, and toast alerts.
- Table Feedback & Sentiment Sync: Stores dining ratings in LocalStorage for manager analytics and customer service tracking.

Future Engineering Scope & Technical Roadmap

Phase 3: Digital Payments & Split-Bill Sync

- Integrated UPI, QR, and credit card payment gateway.
- Automated split-bill share links sent via SMS/WhatsApp.
- Cloud POS synchronization with real-time bill settlements.
- Instant digital invoice generation with QR code.

Phase 4: AI Recommendations & Upselling

- AI dish pairing suggestions based on guest past dining feedback.
- Dynamic daily specials & combo discounts optimization.
- Real-time guest sentiment tracking & kitchen alerts.
- Predictive ingredient restocking alerts based on trends.

Phase 5: Multi-Branch Analytics & Cloud KDS

- Multi-outlet franchise live management dashboard.
- Automated real-time inventory deduction on sales.
- Cross-branch staff allocation and speed-of-service audit.
- Enterprise analytics exportable to Excel, PDF, and cloud ERP.

THANK YOU

RESTAURANT TABLE ORDERING SYSTEM (QUEEN'S KITCHEN)

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GROUP G14 • PROJECT ID: PID-11 • FRONT END ENGINEERING - II (Phase 2)I (Phase 2)I (Phase 2)I